Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A gas expands adiabatically at constant pressure such that $\mathrm{T} \propto \mathrm{V}^{-1/2}$. The value of $\gamma\left(\mathrm{C}_{\mathrm{p,m}} / \mathrm{C}_{\mathrm{vm}}\right)$ of the gas will be :

(A) 1.30

(B) 1.50

(C) 1.70

(D) 2

Q.2

What is the value of change in internal energy at 1 atm in the process?
$\mathrm{H_2O}(1, 323\ \mathrm{K}) \longrightarrow \mathrm{H_2O}(\mathrm{g}, 423\ \mathrm{K})$
Given : $\mathrm{C}_{\mathrm{v,m}}\left(\mathrm{H_2O}, 1\right) = 75.0 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$
$\mathrm{C}_{\mathrm{p,m}}\left(\mathrm{H_2O}, \mathrm{g}\right) = 33.314 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$
$\Delta \mathrm{H}_{\mathrm{vap}}$ at $373\ \mathrm{K} = 40.7\ \mathrm{kJ/mol}$

(A) $42.91\ \mathrm{kJ/mol}$

(B) $43086\ \mathrm{kJ/mol}$

(C) $42.6\ \mathrm{kJ/mol}$

(D) $49.6\ \mathrm{kJ/mol}$

Q.3

An ideal gas expands against a constant external pressure of 2.0 atmosphere from 20 litre to 40 litre and absorbs $10\ \mathrm{kJ}$ of heat from surrounding. What is the change in internal energy of the system? (given : 1 atm-litre $= 101.3\ \mathrm{J}$)

(A) $4052\ \mathrm{J}$

(B) $5948\ \mathrm{J}$

(C) $14052\ \mathrm{J}$

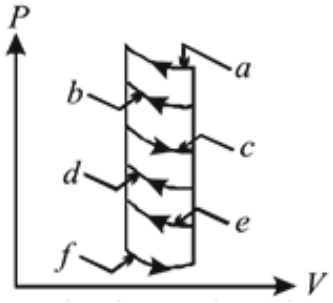
(D) $9940\ \mathrm{J}$

Q.4

The pressure-volume diagram shows six curved paths that can be followed by the gas (connected by vertical

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paths). Which two of them should be part of a closed cycle if the net work done by the gas is to be its maximum positive value?



(A) af

(B) ae

(C) ac

(D) ce

Q.5

On $P-V$ coordinates, the slope of an isothermal curve of a gas at a pressure $P=1\mathrm{MPa}$ and volume $V=0.0025\ \mathrm{m}^3$ is equal to $-400\mathrm{MPa}/\mathrm{m}^3$. If C_p/C_v $=1.4$, the slope of the adiabatic curve passing through this point is :

(A) $-56\mathrm{MPa}/\mathrm{m}^3$

(B) $-400\mathrm{MPa}/\mathrm{m}^3$

(C) $-560\mathrm{MPa}/\mathrm{m}^3$

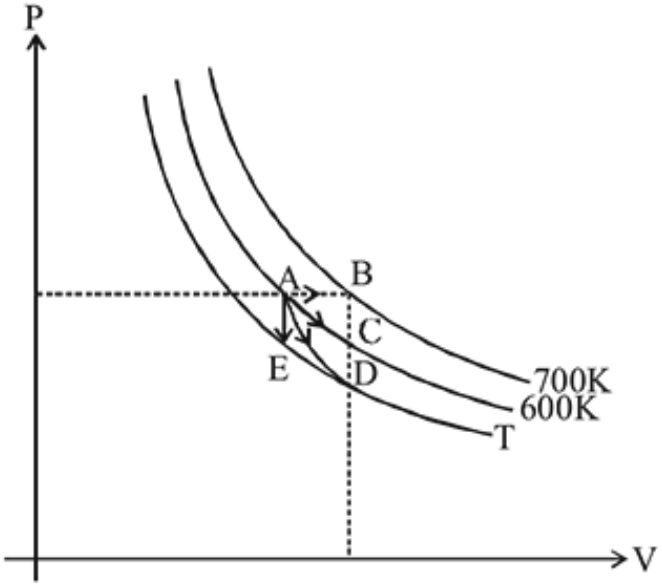
(D) None of these

Q.6

For an ideal gas four processes are marked as 1 2,3 and 4 on $\mathrm{P}-\mathrm{V}$ diagram as shown in figure. The amount of heat supplied to the gas in the process $1,2,3$ and 4 are $\mathrm{Q}_1, \mathrm{Q}_2, \mathrm{Q}_3$ and Q_4 respectively, then correct order of heat supplied to the gas is $[\mathrm{AB}$ is process- $1, \mathrm{AC}$ is process- $2, \mathrm{AD}$ is adiabatic process-3 and AE is

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process-4]



(A) $Q_1 > Q_2 > Q_3 > Q_4$

(B) $Q_1 > Q_2 > Q_4 > Q_3$

(C) $Q_1 > Q_4 > Q_2 > Q_3$

(D) $Q_1 < Q_2 < Q_3 < Q_4$

Q.7

An ideal gas can be expanded from an initial state to a certain volume through two different processes, (A) $PV^2 = K$ and (B) $P = KV^2$, where K is a positive constant. Then, choose the correct option from the following.

(A) Final temperature in (A) will be greater than in (B)

(B) Final temperature in (B) will be greater than in (A)

(C) Work done by the gas in both the processes would be equal

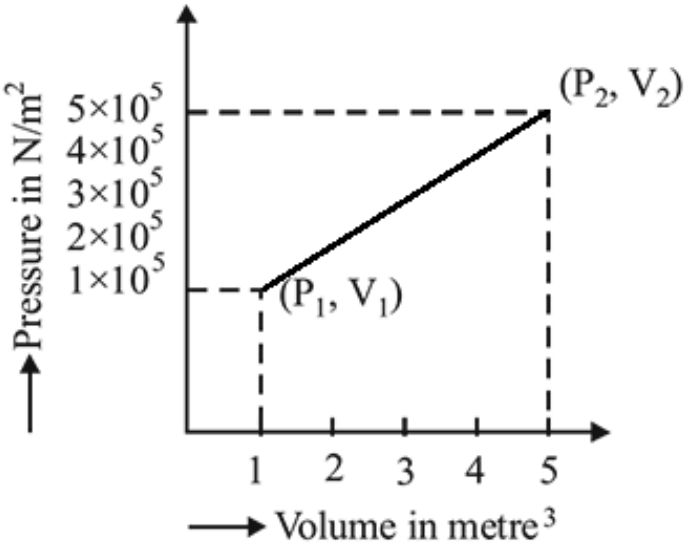
(D) Total heat given to the gas in (A) is greater than in (B)

Q.8

A system changes from the state (P_1, V_1) to $(P_2\ V_2)$ as shown in the figure. What is the work done by the

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system?



(A) 7.5×10^5 joule

(B) 7.5×10^5 erg

(C) 12×10^5 joule

(D) 6×10^5 joule

Q.9

When the state of a gas adiabatically changed from an equilibrium state A to another equilibrium state B an amount of work done on the stystem is $35\ \mathrm{J}$. If the gas is taken from state A to B via process in which the net heat absorbed by the system is $12\mathrm{cal}$, then the net work done by the system is $(1\mathrm{cal} = 4.19\ \mathrm{J})$

(A) $13.2\ \mathrm{J}$

(B) $15.4\ \mathrm{J}$

(C) $12.6\ \mathrm{J}$

(D) $16.8\ \mathrm{J}$

Q.10

An ideal gas is subjected to cyclic process involving four thermodynamic states, the amounts of heat (Q) and work (W) involved in each of these states $Q_1 = 6000\ \mathrm{J}; Q_2 = -5500\ \mathrm{J};\quad Q_3 = -3000\ \mathrm{J}$
$Q_4 = +3500\ \mathrm{J}$
$W_1 = 2500\ \mathrm{J};\quad W_2 = -1000\ \mathrm{J};\quad W_3 = -1200\ \mathrm{J}$
$W_4 = x\ \mathrm{J}$
The ratio of the net work done by the gas to the total heat absorbed by the gas is η. The values of x and η respectively are

(A) $500; 7.5\%$

(B) $700; 10.5\%$

(C) $1000; 21\%$

(D) $1500; 15\%$

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Q.11

For a monoatomic gas, the value of the ratio of $C_{\mathrm{P},m}$ and $C_{v,m}$ is

(A) $5/3$

(B) $7/5$

(C) $9/7$

(D) $9/11$

Q.12

Internal energy of an ideal gas depends on

(A) Volume

(B) Temperature

(C) Pressure

(D) None of these

Q.13

1 mol each of $\mathrm{CaC_2}, \mathrm{Al_4C_3}, \mathrm{Mg_2C_3}$ reacts with $\mathrm{H_2O}$ in separate open flask. Numerical value of work done by the system is in order

(A) $\mathrm{CaC_2} < \mathrm{Al_4C_3} = \mathrm{Mg_2C_3}$

(B) $\mathrm{CaC_2} < \mathrm{Al_4C_3} < \mathrm{Mg_2C_3}$

(C) $\mathrm{CaC_2} = \mathrm{Mg_2C_3} < \mathrm{Al_4C_3}$

(D) $\mathrm{CaC_2} = \mathrm{Mg_2C_3} = \mathrm{A_4C_3}$

Q.14

When the state of a gas adiabatically changed from an equilibrium state A to another equilibrium state B an amount of work done on the stystem is $35\ \mathrm{J}$. If the gas is taken from state A to B via process in which the net heat absorbed by the system is $12\mathrm{cal}$, then the net work done by the system is $(1\mathrm{cal} = 4.19\ \mathrm{J})$

(A) $13.2\ \mathrm{J}$

(B) $15.4\ \mathrm{J}$

(C) $12.6\ \mathrm{J}$

(D) $16.8\ \mathrm{J}$

Q.15

The ratio $\gamma = C_{p,\,\mathrm{m}}/C_{V,\,\mathrm{m}}$ for a diatomic gaseous molecules is

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(A) 1.667

(B) 1.400

(C) 1.154

(D) 1.167

Answers & Solutions

Q.1 Answer:

1.50

Solution:

$\mathrm{T} \propto \frac{1}{\sqrt{\mathrm{V}}} ; \mathrm{TV}^{1/2}=\text{constant}$

For adiabatic process, $\mathrm{TV}^{\gamma-1}=$ **constant**

$\therefore \gamma-1=\frac{1}{2}, \gamma=\frac{3}{2}$

Q.2 Answer:

42.6 kJ/mol

Solution:

Given,

$\mathrm{H_2O}(\ell, 323\ \mathrm{K}) \rightarrow \mathrm{H_2O}\ (9, \mathrm{H23\ K})$

$\mathrm{Cv, m}\ (\mathrm{H_2O}, \ell)=75.0 \mathrm{J\ k^{-1}\ mol^{-1}}$

$\mathrm{Cp, m}\ (\mathrm{H_2O}, \mathrm{g})=33.31 \mathrm{H\ K\ K^{-1}\ mol^{-1}}$

$\Delta \mathrm{H}$ vap at $373\ \mathrm{K}=40.7\ \mathrm{kJ} / \mathrm{mol}^{-1}$

The steps involved in the above conversion is

$\mathrm{H_2O}(\ell, 323\ \mathrm{K}) \underset{\Delta \mathrm{U_1}}{\longrightarrow} \mathrm{H_2O}(\ell, 373\ \mathrm{K}) \underset{\Delta \mathrm{U_2}}{\longrightarrow} \mathrm{H_2O}(9, 373\ \mathrm{K}) \underset{\Delta \mathrm{U_3}}{\longrightarrow} \mathrm{H_2O}(9, \mathrm{H23k})$

$\Delta U_{\text{total }}=\Delta U_1+\Delta U_2+\Delta U_3$

$\Delta U_1=\mathrm{Cv, m}\ (H_2O, \ell) \times \Delta T=75 \times 50 \times 10^{-3}\ \mathrm{kJ/mol}$

$\Delta U_2=\Delta H-\Delta D R T=40.7-1 \times 8.314 \times 373 \times 10^{-3}$

$=37.598\ \mathrm{kJ/mol}$

$\Delta U_3=(v, m\ (H_2\mathrm{O}, g)\ \Delta T=(Cp, m\ (\mathrm{H_2O}, g)-\mathrm{R})\ \Delta T$

$=(33.314-8.314)(423-373) \times 10^{-3}$

$=25 \times 50 \times 10^{-3}\ \mathrm{kJ/mol}$

$\Delta U=75 \times 50 \times 10^{-3}+37.598+25 \times 50 \times 10^{-3}\ \mathrm{kJ} \mid \mathrm{mol}$

$\Delta U=42.598\ \mathrm{kJ/mol}=42.6\ \mathrm{kJ/mol}$

Q.3 Answer:

5948 J

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Solution:

$\Delta U = q + w$
$= 10 \times 1000 - 2 \times (20) \times 101.3 = 5948 \text{ J}$

Q.4 Answer:

af

Solution:

Work done in expansion is positive and in compression it is negative.

Q.5 Answer:

-560MPa/m^3

Solution:

Slope of adiabatic curve $= \gamma \times$ **slope of isothermal curve** $= 1.4 \times (-400) = -560 \text{MPa/m}^3$

Q.6 Answer:

$Q_1 > Q_2 > Q_3 > Q_4$

Solution:

In process- 1 heat supplied $=$ **area under** AB **curve** $+ n \times c_p \times 100$ **(isochoric process) In process-2 heat supplied** $=$ **area under** AC **curve (isothermal process) In process-3 heat supplied** $= 0$ **(adiabatic process) In process- 4 heat supplied** $= n \times c_v (T - 600)$
(isobaric process)

Q.7 Answer:

Final temperature in (B) will be greater than in (A)

Solution:

Process
(A) $\frac{nRT}{V} V^2 = K; TV = \text{constant};$
$T \propto \frac{1}{V}$
Process (B) $\frac{nRT}{V} = KV^2; \frac{T}{V^3} =$ **constant;**
$T \propto V^3$

Q.8 Answer:

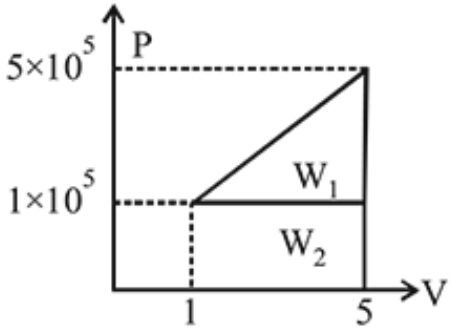
12×10^5 joule

Solution:

$W = W_1 + W_2$
$= 4 \times 1 \times 10^5 + 1/2 \times 4 \times 10^5 \times 4$

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$= 12 \times 10^5$ **Joule**



Q.9 Answer:

15.4 J

Solution:

In the first-case adiabatic change, $\Delta Q = 0, \Delta W = -35$ J

From 1st **law of thermodynamics,** $\Delta Q = \Delta U + \Delta W$

or $0 = \Delta U - 35$

$\therefore \Delta U = 35$ J

In the second case $\Delta Q = 12\text{cal} = 12 \times 4.2$ J $= 50.4$ J

$\Delta W = \Delta Q - \Delta U = 50.4 - 35 = 15.4$ J

Q.10 Answer:

$700; 10.5\%$

Solution:

$Q = Q_1 + Q_2 + Q_3 + Q_4$

$= 6000 - 5500 - 3000 + 3500 = +1000$ J

$W = W_1 + W_2 + W_3 + W_4$

$= 2500 - 1000 - 1200 + x = +300 + x$

In cyclic process, $\Delta U = 0$ **Now,** $Q = \Delta U + W$

or $1000 = 0 + (300 + x)$

$\therefore x = 700$ J

$\eta = \frac{W}{Q_1 + Q_4}$

$= \frac{1000}{6000+3500} = 10.5\%$

Q.11 Answer:

$5/3$

Solution:

For a monoatomic gas, $C_{V,m} = (3/2)R$ and $C_{p,m} = (5/2)R$. Hence $C_{p,m}/C_{V,m} = 5/3$.

Q.12 Answer:

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Temperature

Solution:

In an ideal gas the inter-molecular collisions are assumed to be absent and the collisions are perfectly elastic. Thus, the gas possesses only translational kinetic energy and hence the internal energy of the ideal gas depends only on temperature.

Q.13 Answer:

$\mathrm{CaC_2} = \mathrm{Mg_2C_3} < \mathrm{Al_4C_3}$

Solution:

$\mathrm{CaC_{2ml}s} + 2\mathrm{H_2O} \rightarrow \mathrm{CaOH_2} + \mathrm{C_2H_2\ g}$

$\mathrm{Al_4C_3\ s} + 12\mathrm{H_2O} \rightarrow 4\mathrm{AlOH_3} + 3\mathrm{CH_4\ g}$

$\underset{\text{lmel}}{\mathbf{Mg_2C_3s}} + 4\mathrm{H_2O} \rightarrow 2\mathrm{MgOH_2} + \overset{3\text{ mol}}{\mathrm{C_3H_4\ g}}$

Thus, $\mathrm{CaC_2} = \mathrm{Mg_2C_3} < \mathrm{Al_4C_3}$

Thus, (c)

Q.14 Answer:

$15.4\ \mathrm{J}$

Solution:

In the first-case adiabatic change, $\Delta \mathrm{Q} = 0, \Delta \mathrm{W} = -35\ \mathrm{J}$

From 1st **law of thermodynamics,** $\Delta \mathrm{Q} = \Delta \mathrm{U} + \Delta \mathrm{W}$

or $\quad 0 = \Delta \mathrm{U} - 35$

$\therefore \Delta \mathrm{U} = 35\ \mathrm{J}$

In the second case $\Delta \mathrm{Q} = 12\mathrm{cal} = 12 \times 4.2\ \mathrm{J} = 50.4\ \mathrm{J}$

$\Delta \mathrm{W} = \Delta \mathrm{Q} - \Delta \mathrm{U} = 50.4 - 35 = 15.4\ \mathrm{J}$

Q.15 Answer:

1.400

Solution:

To find:

$\frac{c_p}{c_v}$ **for diatomic gas.**

$C_p \rightarrow$ **Molar specific heat capacity at constant pressure.**

$C_v \rightarrow$ **Molar specific heat capacity at constant volume..**

$\frac{c_p}{c_v} = 1 + \frac{2}{f}$

$f \rightarrow$ **degree of freedom of gas molecules.**

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A molecule of diatomic gas has 5 degree of freedom, i.e., $f = 5$.

$$\therefore \frac{C_p}{C_v} = 1 + \frac{2}{5} = \frac{7}{5} = 1.4$$

$$\therefore \frac{C_p}{C_v} \text{ for diatomic gas } = 1.4$$

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